Introduction to VLSI Design
2024 Homework 2 — Worked Solutions

Independent derivation with reference-answer review · 13 October 2026

Scope. The supplied 2024 Homework 2 contains two analytical problems. The answers below use its stated constant-threshold, constant-leakage and constant-on-resistance assumptions. Supplementary HSPICE decks are supplied separately; the CIC BSIM3 model must not be confused with those assumptions.

1. Non-ideal effect

(a) Precharge voltage

For Pre = 0, the PMOS connects BL to V_{dd} without an NMOS-style high-level threshold loss. For SL = 1, take the NMOS gate to be V_{dd}. As the storage-node voltage V_C rises,

$$V_{GS,n} = V_{\mathrm{dd}} - V_C.$$

In the question's threshold-cutoff approximation, charging stops at $V_{GS,n} = V_t$. Therefore

$$\boxed{V_C(0) = V_{\mathrm{dd}} - V_t.} \tag{1}$$

The threshold loss is across the access NMOS, not the precharge PMOS. Here V_t is the threshold stipulated by the question. If body effect is included, it becomes an implicit equation $V_C = V_{\mathrm{dd}} - V_t(V_{SB} = V_C)$ rather than a subtraction using the zero-body-bias threshold. Subthreshold conduction also removes an ideal abrupt cutoff.

(b) Stored-data retention time

Let $t = 0$ be the end of precharge. Assume that the storage-side NMOS junction removes a constant current $I_L > 0$ from the capacitor, with all other storage-node leakage and loading neglected. Charge conservation gives

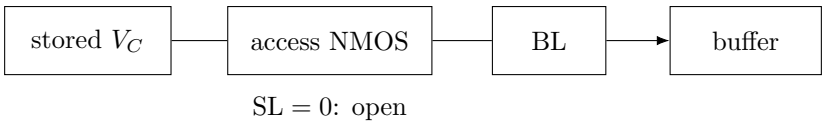
$$Q = CV_C, \qquad C\frac{dV_C}{dt} = -I_L, \qquad V_C(t) = V_{\mathrm{dd}} - V_t - \frac{I_L}{C}t.$$

The boundary at which a stored high can no longer exceed the stated buffer threshold is $V_C(t_{\mathrm{ret}}) = 0.4V_{\mathrm{dd}}$. Hence

$$\boxed{t_{\mathrm{ret}} = \frac{C(0.6V_{\mathrm{dd}} - V_t)}{I_L}, \quad \text{provided } V_t < 0.6V_{\mathrm{dd}}.} \tag{2}$$

The data is safely high for $0 \le t < t_{\mathrm{ret}}$ in this ideal model; equality is the switching boundary, with no noise margin. If $V_t \ge 0.6V_{\mathrm{dd}}$, the initial stored voltage does not exceed the threshold, so there is no positive guaranteed high-retention interval. If $I_L = 0$ and the initial margin is positive, this ideal model predicts infinite retention. No numerical C, V_t, V_{dd} or I_L is specified by the assignment.

Important topology qualification. In the original Fig. 1, the buffer input is attached to BL, while C is on the other side of the access NMOS. When SL is low, these are separate nodes:



SL = 0: open

Equation (2) is the intended *stored-charge* answer, under an implicit ideal read that exposes the storage voltage to the buffer without appreciable disturbance. It is not a prediction of continuously observed Out while both transistors remain off. That literal interpretation needs BL capacitance, BL leakage and a read protocol, none of which are given. For example, upon reselecting a cell with BL capacitance C_{BL} initially at $V_{BL,0}$, ideal charge sharing would give

$$V_{\text{share}} = \frac{CV_C + C_{BL}V_{BL,0}}{C + C_{BL}},$$

before considering threshold loss or buffer loading. The threshold condition must then be imposed on the actual read voltage. No extra factor of two is applied to I_L: only the storage-connected junction directly discharges C while the access transistor is off.

Illustrative calculation, not assignment data. Choosing $V_{\text{dd}} = 1.8\,\text{V}$, $V_t = 0.45\,\text{V}$, $C = 20\,\text{fF}$ and $I_L = 1\,\text{pA}$ gives

$$V_C(0) = 1.35\,\text{V}, \quad V_{\text{switch}} = 0.72\,\text{V}, \quad t_{\text{ret}} = 12.6\,\text{ms}.$$

The deck `q1_constant_leakage.sp` implements this ideal capacitor and current-source model directly, not the full access-transistor circuit.

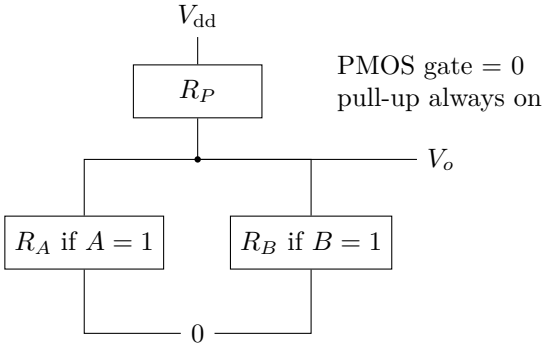
2. Logic gate

(a) Boolean function

The PMOS gate is tied to ground, so the pull-up remains on. The two NMOS devices form parallel pull-down paths. Either high input pulls the output low, while both inputs low leave the output high. Thus the intended logic is

$$\boxed{Y = \overline{A + B} = \bar{A}\,\bar{B}, \quad \text{a pseudo-NMOS NOR gate.}} \tag{3}$$

It is ratioed logic: correct low levels require sufficiently strong NMOS devices, and low-output states draw static supply current.



(b) Transistor sizing

The phrase "smallest voltage of logic 0 is $0.2V_{\text{dd}}$" is interpreted as the *maximum acceptable logic-low voltage*, $V_{OL,\max} = 0.2V_{\text{dd}}$. A smallest allowed low voltage would not define a sensible sizing requirement, because ground is also a valid low. This interpretation is also used in the provided handwritten answer.

Let S_0 be an arbitrary reference aspect ratio, and define

$$s_A = \frac{(W/L)_{nA}}{S_0}, \quad s_B = \frac{(W/L)_{nB}}{S_0}, \quad s_p = \frac{(W/L)_p}{S_0}.$$

For equal reference sizes the assignment specifies $R_p = 2R_n$. With resistance inversely proportional to aspect ratio,

$$R_A = \frac{R_n}{s_A}, \qquad R_B = \frac{R_n}{s_B}, \qquad R_P = \frac{2R_n}{s_p}.$$

For $A = 1, B = 0$, the constant-resistance divider gives

$$\frac{V_{10}}{V_{\mathrm{dd}}} = \frac{R_A}{R_P + R_A} = \frac{s_p}{s_p + 2s_A} \le 0.2 \quad \Longrightarrow \quad s_A \ge 2s_p.$$

The $A = 0, B = 1$ case independently requires $s_B \ge 2s_p$. Therefore

$$\boxed{(W/L)_{nA} \ge 2\,(W/L)_p\,, \qquad (W/L)_{nB} \ge 2\,(W/L)_p\,.} \tag{4}$$

The minimum symmetric ratios are $(W/L)_{nA} = (W/L)_{nB} = 2(W/L)_p$. For equal channel lengths, each NMOS width is at least twice the PMOS width. The assignment fixes ratios, not unique absolute widths or lengths.

At the boundary, write $R_A = R_B = R$ and $R_P = 4R$. The complete truth table is

A	B	Pull-down resistance	V_o/V_{dd}	Logic
0	0	open	1	1
0	1	R	$1/5 = 0.2$	0
1	0	R	$1/5 = 0.2$	0
1	1	$R/2$	$1/9 \simeq 0.1111$	0

The worst low occurs with only one NMOS on; sizing from the two-on case would be insufficient. The equality solution has no additional margin to the stated $0.2V_{\mathrm{dd}}$ limit. Larger NMOS-to-PMOS ratios reduce the low voltage in this resistor model, at the cost of more input capacitance and area.

Numerical resistor check. For $V_{\mathrm{dd}} = 1.8\,\mathrm{V}$, $R = 1\,\mathrm{k\Omega}$, $R_P = 4\,\mathrm{k\Omega}$:

$$(V_{00}, V_{01}, V_{10}, V_{11}) = (1.8, 0.36, 0.36, 0.20)\,\mathrm{V}.$$

The one-on supply current is $1.8/(4000 + 1000) = 0.36\,\mathrm{mA}$; the corresponding total static power is $0.648\,\mathrm{mW}$. These absolute resistor values are illustrative and are not extracted from CIC.

Supplementary numerical validation and reproducibility

File	Purpose
`q1_constant_leakage.sp`	Constant leakage and initial-voltage abstraction; measures threshold-crossing time.
`q2_constant_resistance.sp`	Four exact resistor-divider input cases.
`q2_cic018_tt.sp`	BSIM3 pseudo-NMOS NOR, TT, 25°C, 1.8 V; sweeps NMOS/PMOS aspect-ratio ratios 1, 2, 3, 4, 6 and 8.
`reproduce.py`	Runs all decks, records exit status and raw console logs; separately writes closed-form expectations.

The CIC deck uses $L_n = L_p = 0.18\,\mu\mathrm{m}$, $W_p = 0.5\,\mu\mathrm{m}$, and $W_n = rW_p$. It measures all four DC-settled output voltages and supply currents. A valid numerical low must satisfy $V_{01}, V_{10}, V_{11} \le 0.36$ V at this chosen supply; a small output number alone does not verify the high state. The voltage-source currents are negated to report positive current delivered by each supply. Ideal voltage inputs are used and output loading is zero.

BSIM3 has voltage-dependent currents, body effects and subthreshold current; it does not impose the problem's constant $R_p = 2R_n$ rule. Consequently, a BSIM3 voltage at ratio two need not equal $0.2V_{\mathrm{dd}}$ and cannot replace the analytical answer in Eq. (4). The local library is included in a protected section to avoid expanding it into a public listing; the model file itself is not part of this submission. The model is identified locally as CIC 0.18 μm, BSIM3, version 1.3, 21 January 2008. For BSIM3 model physics, see the UC Berkeley BSIM3v3.2 manual.

Actual execution. All three decks completed successfully with PrimeSim HSPICE S-2021.09 on 13 October 2026 (exit code 0). The measured ideal retention is 12.600025 ms, versus the closed-form 12.600000 ms: relative error 1.98×10^{-6}, due to transient startup and integration. At 12.6 ms the measured voltage is 0.72000125 V. The ideal resistor deck measures exactly the four voltages listed above to its reported precision.

The following are actual CIC TT measurements at 25°C:

$r = W_n/W_p$	V_{00} (V)	$V_{01} = V_{10}$ (V)	V_{11} (V)	P_{01} (mW)
1	1.800000	0.176482	0.078389	0.242883
2	1.800000	0.078122	0.036908	0.245440
3	1.800000	0.049900	0.024040	0.246127
4	1.800000	0.036566	0.017786	0.246445
6	1.800000	0.023779	0.011676	0.246746
8	1.800000	0.017604	0.008684	0.246891

For example, at ratio two the measured one-on supply current is 0.1363557 mA. Even ratio one passes the low-voltage criterion for this particular TT model and bias; this neither changes the assignment's constant-resistance requirement nor establishes a minimum ratio across process corners, voltage and temperature.

The files `*_measurements.csv` are parsed from HSPICE measurement tables, while `analytic_expected.csv` contains only closed-form calculations. `simulation_status.csv` and `validation_summary.json` record successful execution without local license metadata. Run `python reproduce.py` to repeat the simulations; raw simulator files go to a temporary local directory by default. `validate_results.py` compares the ideal results to the equations, checks all sampled CIC output levels and symmetry, and verifies static power against supply current. All 43 numerical checks passed; their observations and tolerances are in `validation_checks.csv`.

Reference-answer comparison

The provided handwritten `VLSI_HW2.pdf` was reviewed after deriving the solution independently. Its precharge and retention formulas agree with Eqs. (1)–(2), and its NOR identification, $R_P = 4R_N$ boundary, size ratio two, and $V_{11} = V_{\mathrm{dd}}/9$ agree with the constant-resistance derivation. Those results are retained.

The final answer adds the initial-readability condition, the BL isolation qualification, the general size inequality, and the distinction between ideal resistors and a BSIM3 device calculation. The handwritten text near the start of part 2(a) says the NMOS turns on for "$A = 1 \wedge B = 1$"; for the parallel network, either individual NMOS turns on with its own high input. The subsequent truth-case analysis in that answer correctly uses the OR condition. No numerical HSPICE results appear in that reference, and none have been inferred from it.